



Master 1 EEA-SIA2

Signal, Image et Apprentissage Automatique

Bureau d'études Apprentissage automatique
(machine learning)

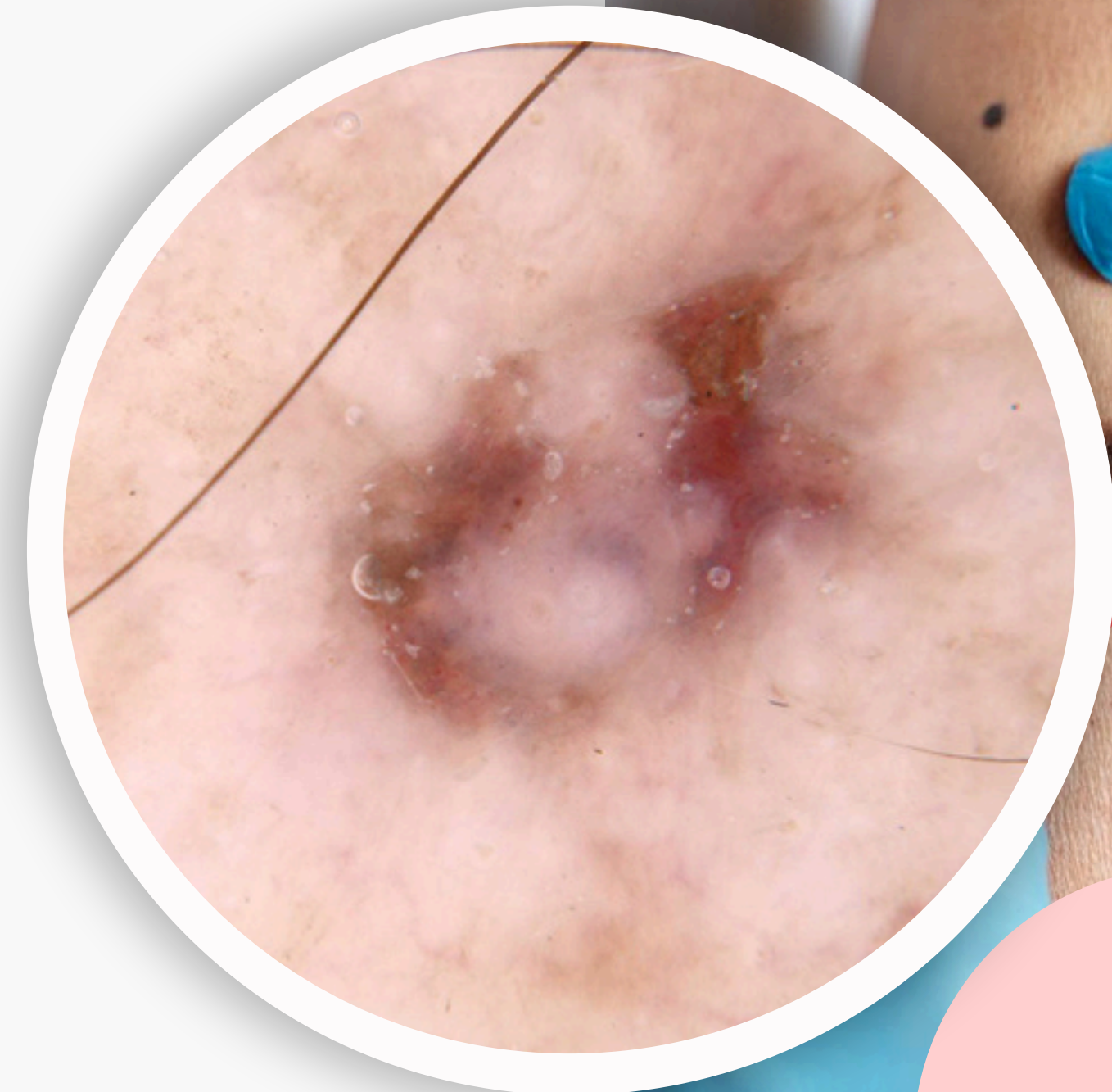
DIAGNOSTIC DU CANCER DE LA PEAU

Réalisé par:

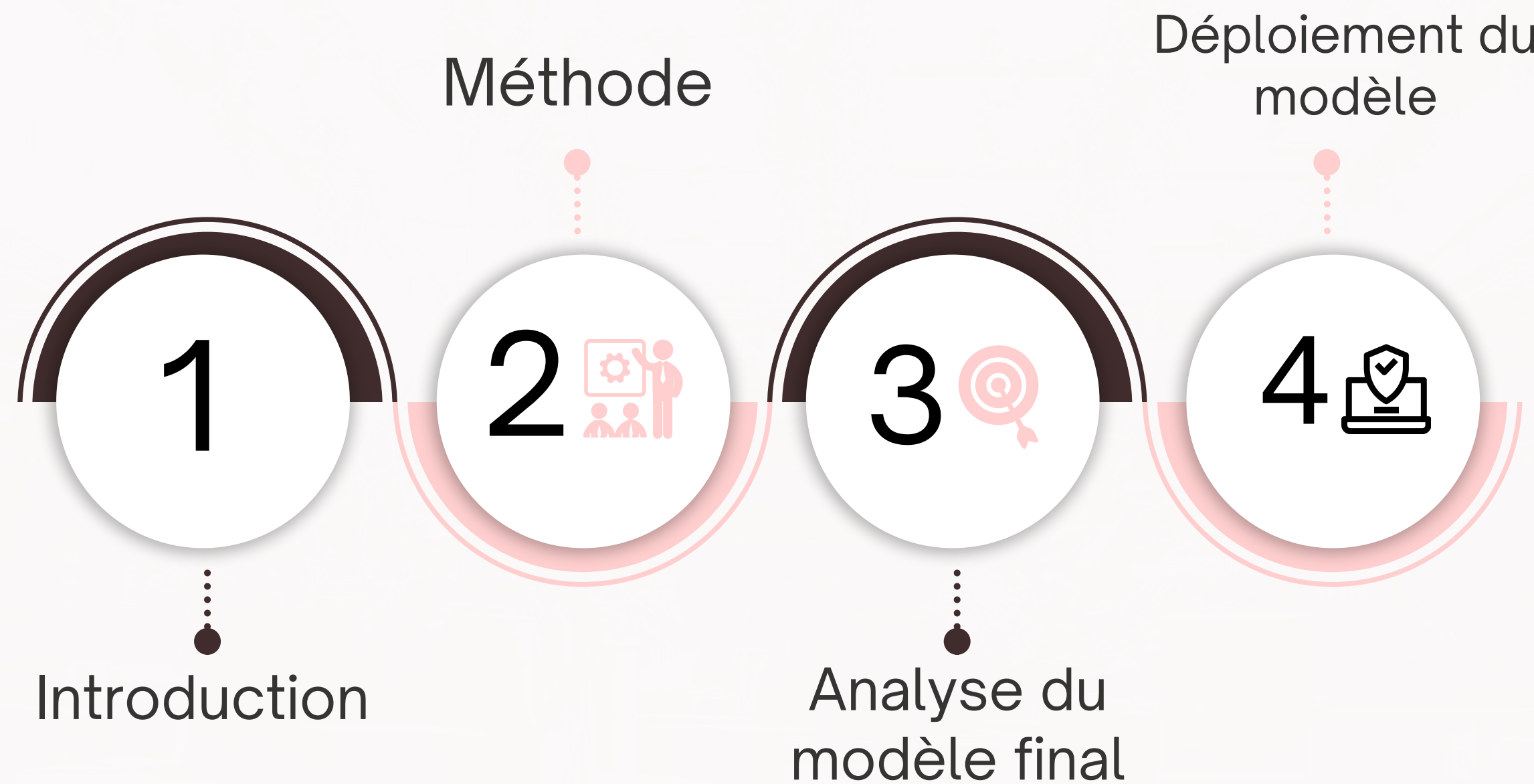
François AWOUNANG

Pavidot DONFACK

Merveille KABAWA



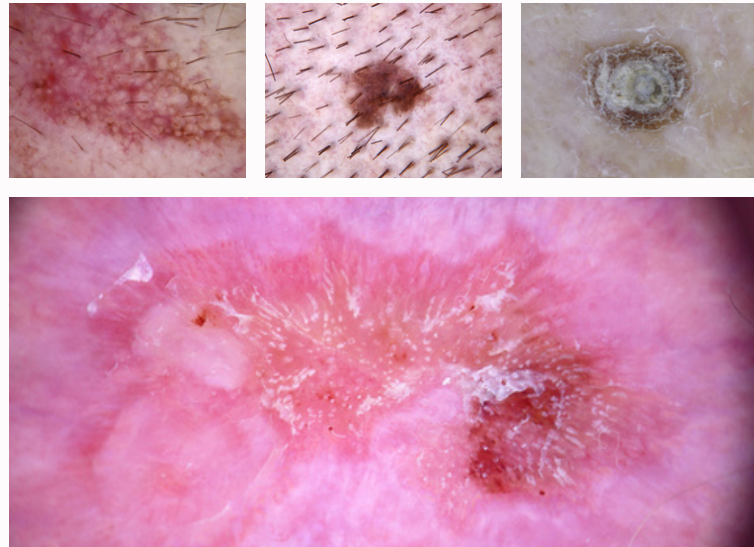
Plan de la présentation



Introduction

► Problématique

Fig: Image dermatoscopique de lésion de type actinic keratosis

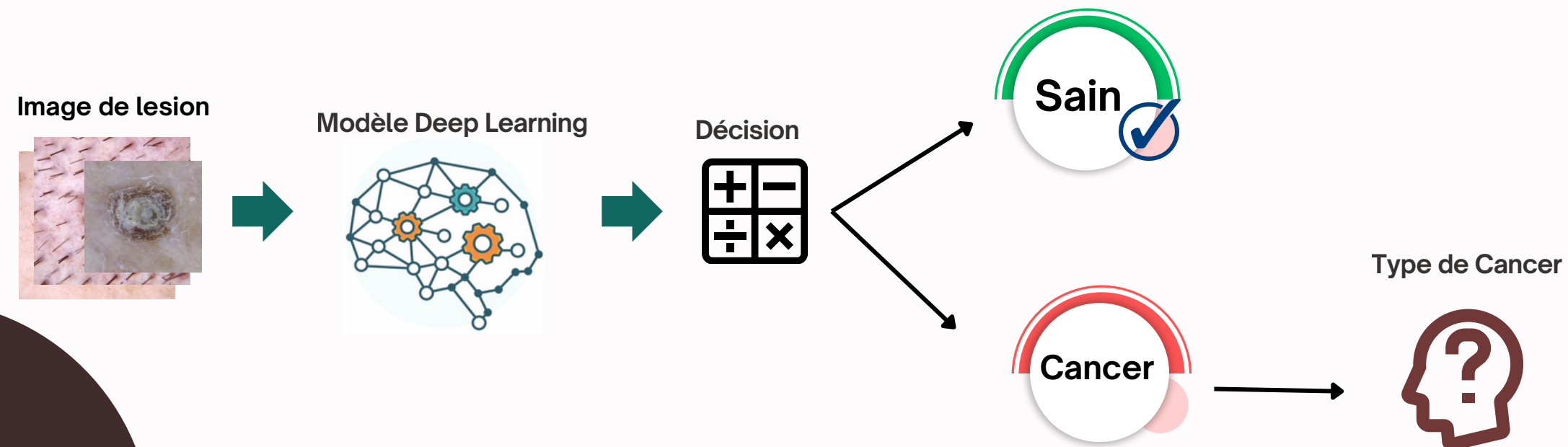


**Besoin
d'automatisation pour
aider au diagnostic**



► Cahier des charges

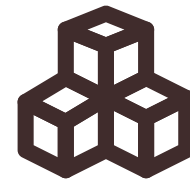
Application



Diagnostic visuel difficile

Methodes

Datasets et Pretraitements



Ici nous allons faire une presentation du dataset , de ses specificites et des differentes operation de pretraitement qui ont ete effectuees ainsi que leurs raisons d’etre.



Architecture et Entraînement



La construction dy modeles c’est a dire le choix du type de models et des la constructions des differentes couches du modele, et aussi de la logique d’entrainement pour tirer le maximum du model.



Validation et modèle de décision

Deploiement

Datasets et Pretraitements

Présentation du dataset

Skin cancer ISIC The International Skin Imaging
Collaboration

Structure du Dataset

DATA

Test

actinic keratosis

basal cell carcinoma

dermatofibroma

melanoma

nevus

pigmented benign keratosis

seborrheic keratosis

squamous cell carcinoma

vascular lesion

Train

actinic keratosis

basal cell carcinoma

dermatofibroma

melanoma

nevus

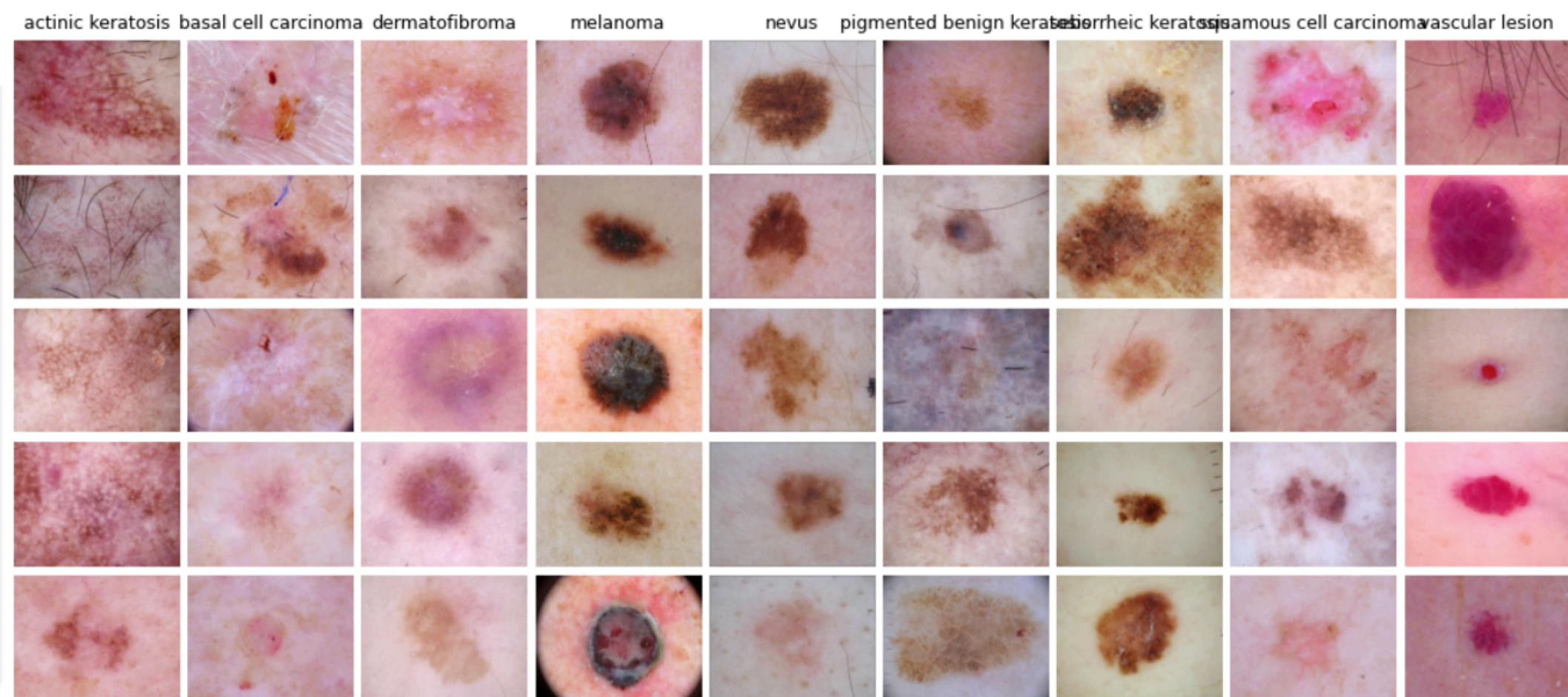
pigmented benign keratosis

seborrheic keratosis

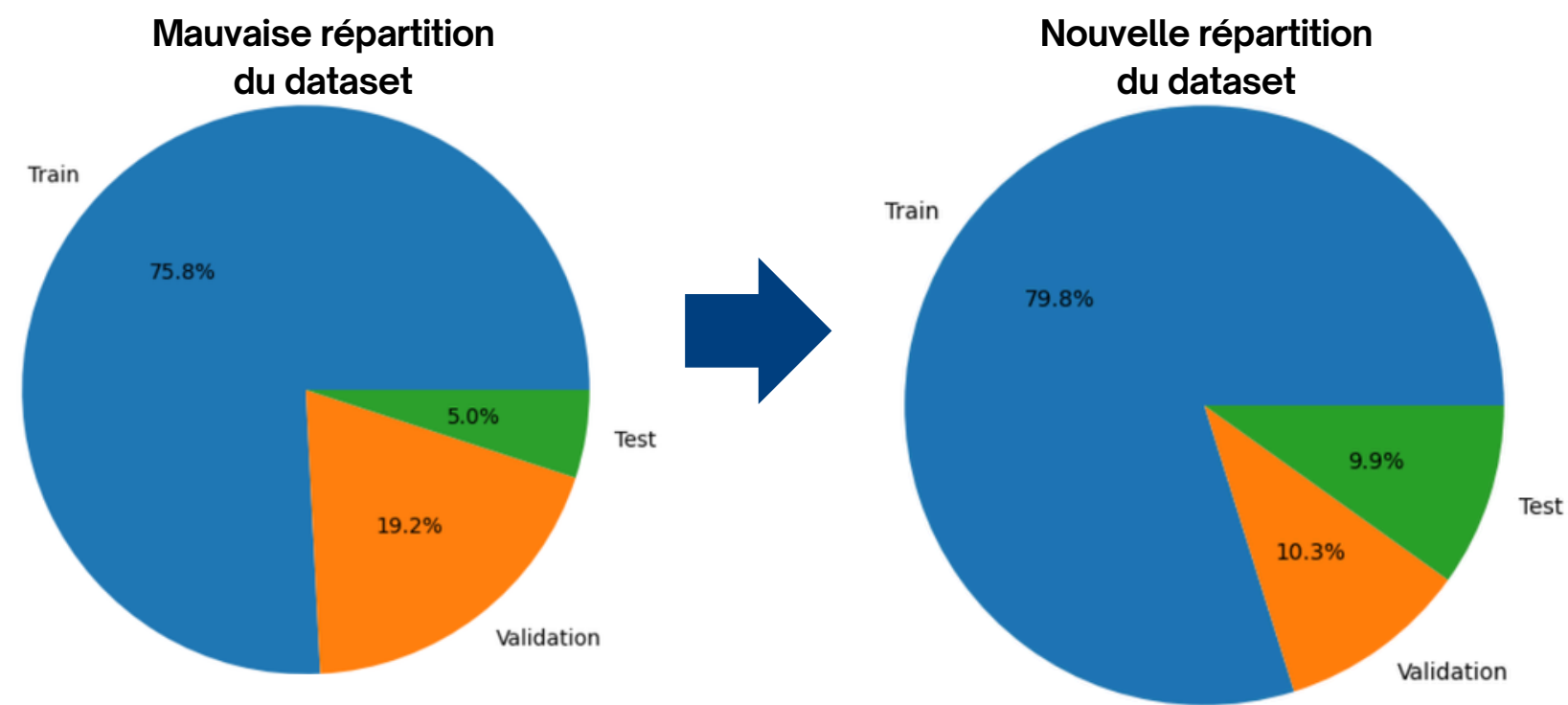
squamous cell carcinoma

vascular lesion

Visualisation des 5 premières images dans chaque classe



Datasets et Pretraitements

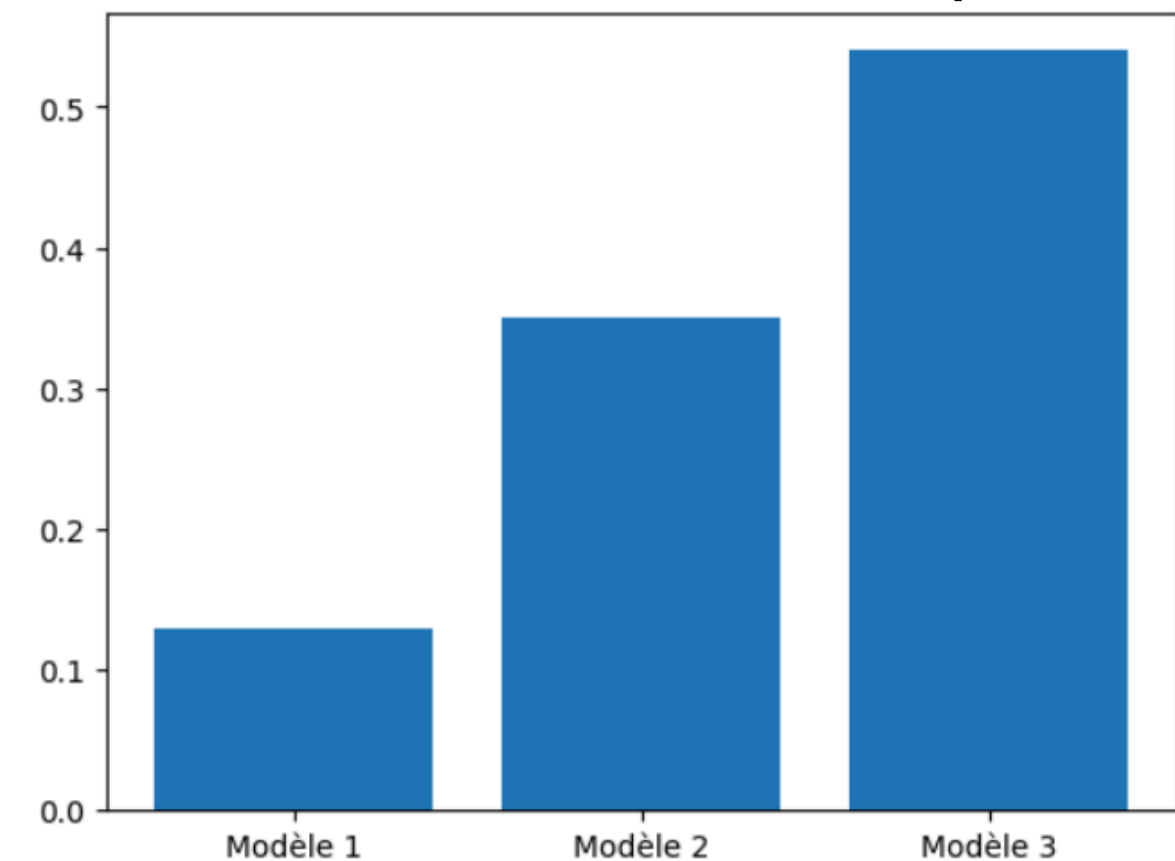


Construction modèle 2 et 3

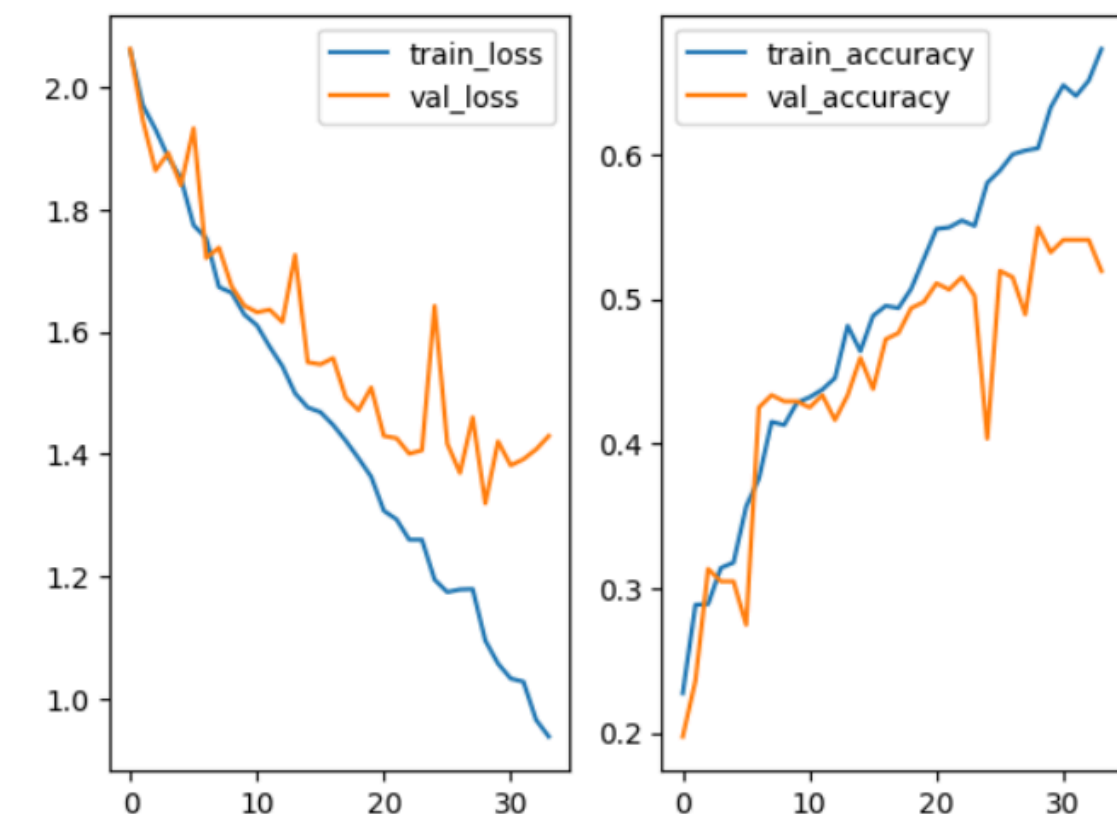
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 224, 224, 32)	896
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
conv2d_1 (Conv2D)	(None, 112, 112, 50)	14,450
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 50)	0
flatten (Flatten)	(None, 156800)	0
dense (Dense)	(None, 500)	78,400,500
dropout (Dropout)	(None, 500)	0
dense_1 (Dense)	(None, 9)	4,509

Total params: 78,420,355 (299.15 MB)

Evolution des modèles simple



8/8 4s 502ms/step - accuracy: 0.5413 - loss: 1.3251
Test accuracy: 0.5413222908973694



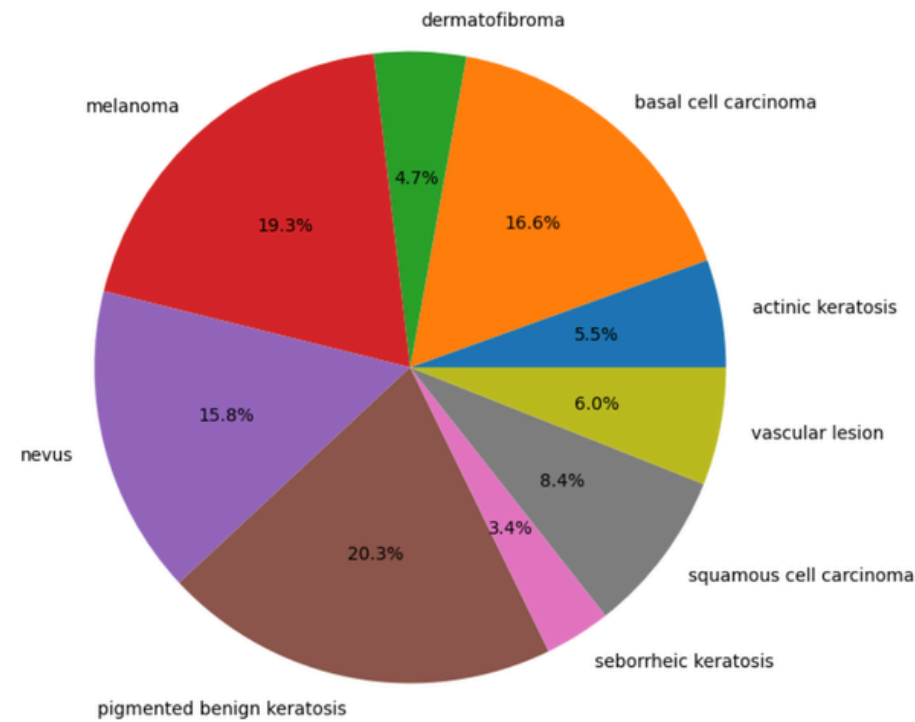
Methodes

Modèle 4: Transfert Learning avec VGG16

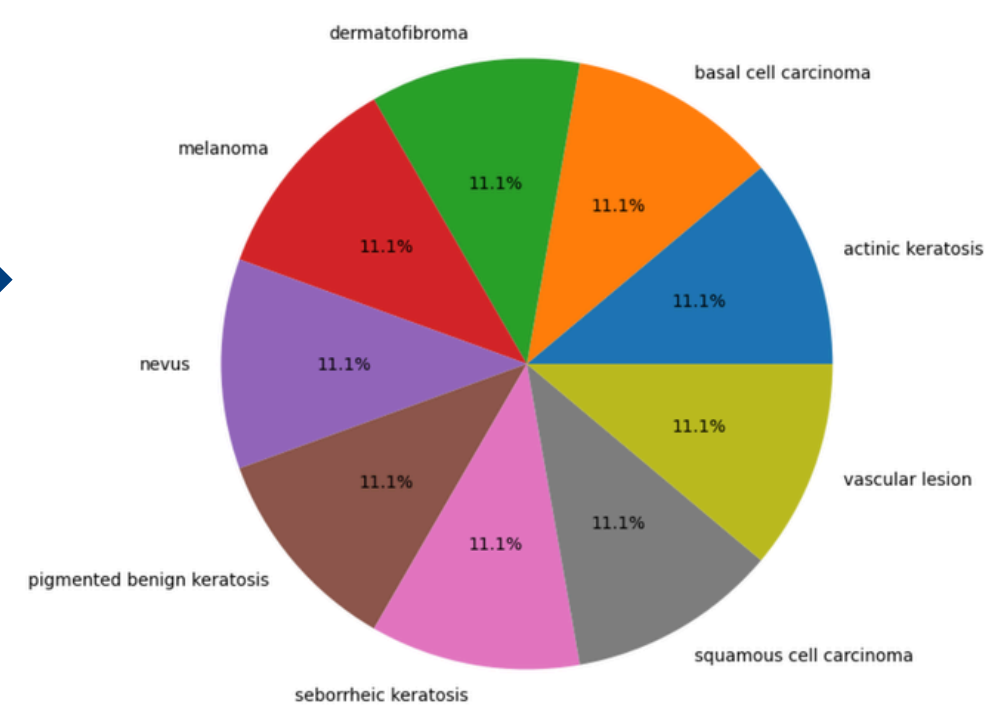
Modèle 5: Affinage du modèle VGG16

Modèle 6 : modèle plus profond, ajout de couches denses

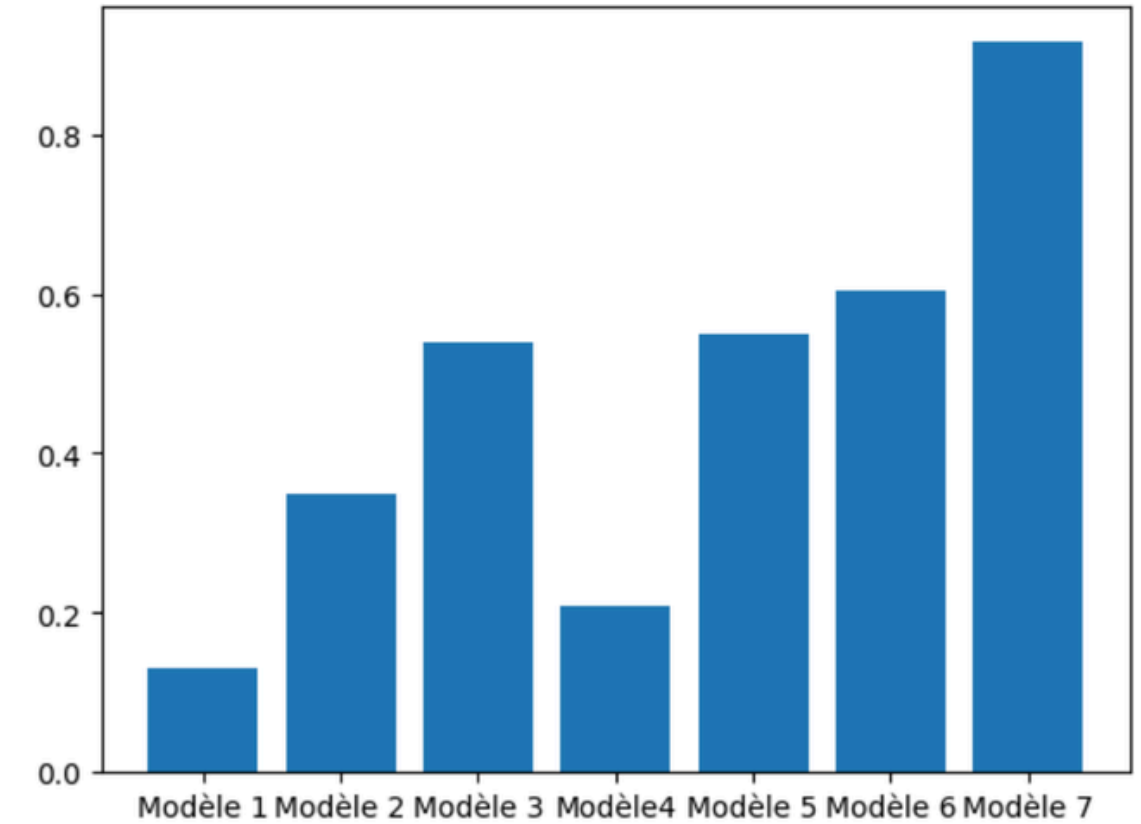
Problème avec la base de données



Solution : data augmentation et rééquilibrage des classes

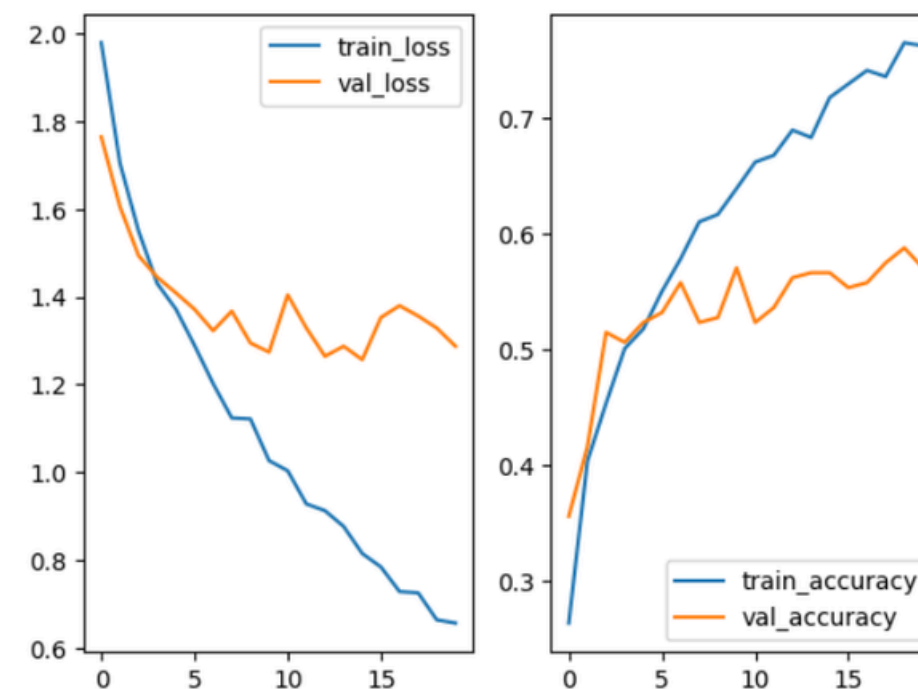


Evolution des modèles



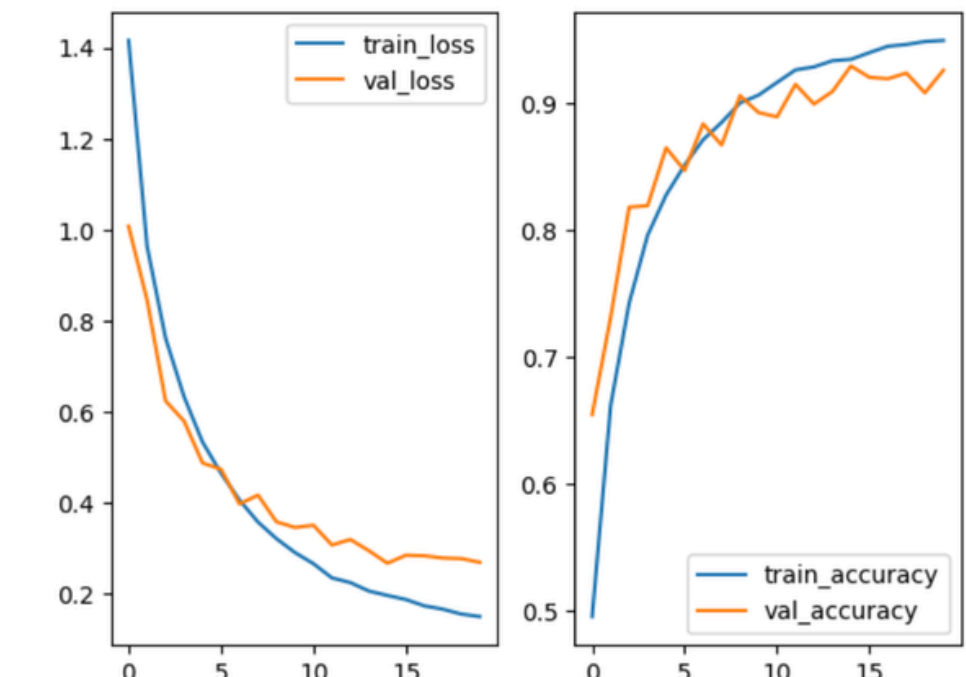
Modèle 6 : modèle plus profond, ajout de couches denses

8/8 5s 557ms/step - accuracy: 0.6033 - loss: 1.1710
Test accuracy: 0.6033057570457458



Modèle 7 : Fine tuning avec les données augmentées

29/29 9s 315ms/step - accuracy: 0.9156 - loss: 0.2888
Test accuracy: 0.915555367469788



Modèle 7: Fine tuning avec les données augmentées

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 7, 7, 512)	14,714,688
flatten_6 (Flatten)	(None, 25088)	0
dense_13 (Dense)	(None, 500)	12,544,500
dropout_8 (Dropout)	(None, 500)	0
dense_14 (Dense)	(None, 9)	4,509

Total params: 27,263,699 (104.00 MB)
Trainable params: 14,908,817 (56.87 MB)
Non-trainable params: 12,354,880 (47.13 MB)

modèl de base de notre modèle 7

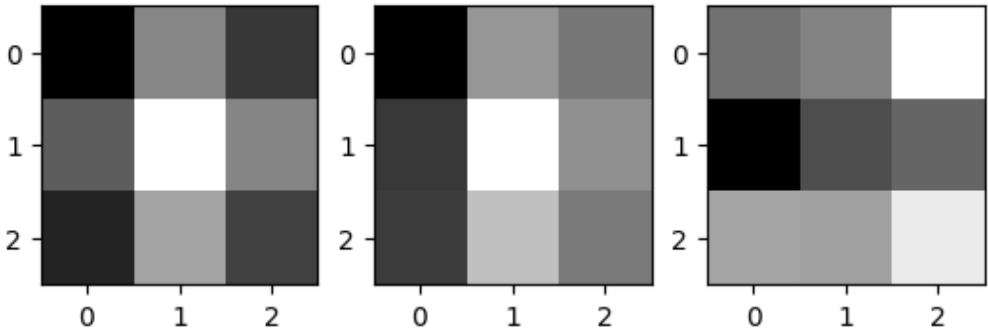
Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_3 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0

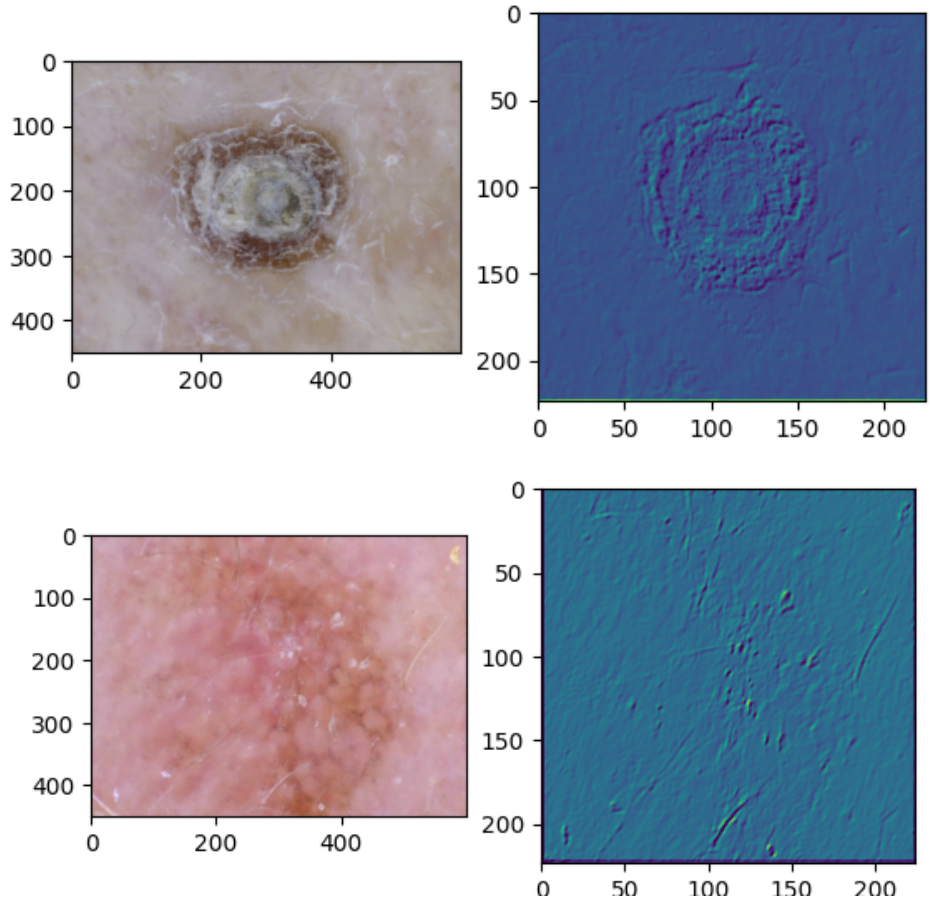
Total params: 14,714,688 (56.13 MB)
Trainable params: 2,359,808 (9.00 MB)
Non-trainable params: 12,354,880 (47.13 MB)

Visualisation des filtres, des
caracteristiques extraites et la matrice de
confusion

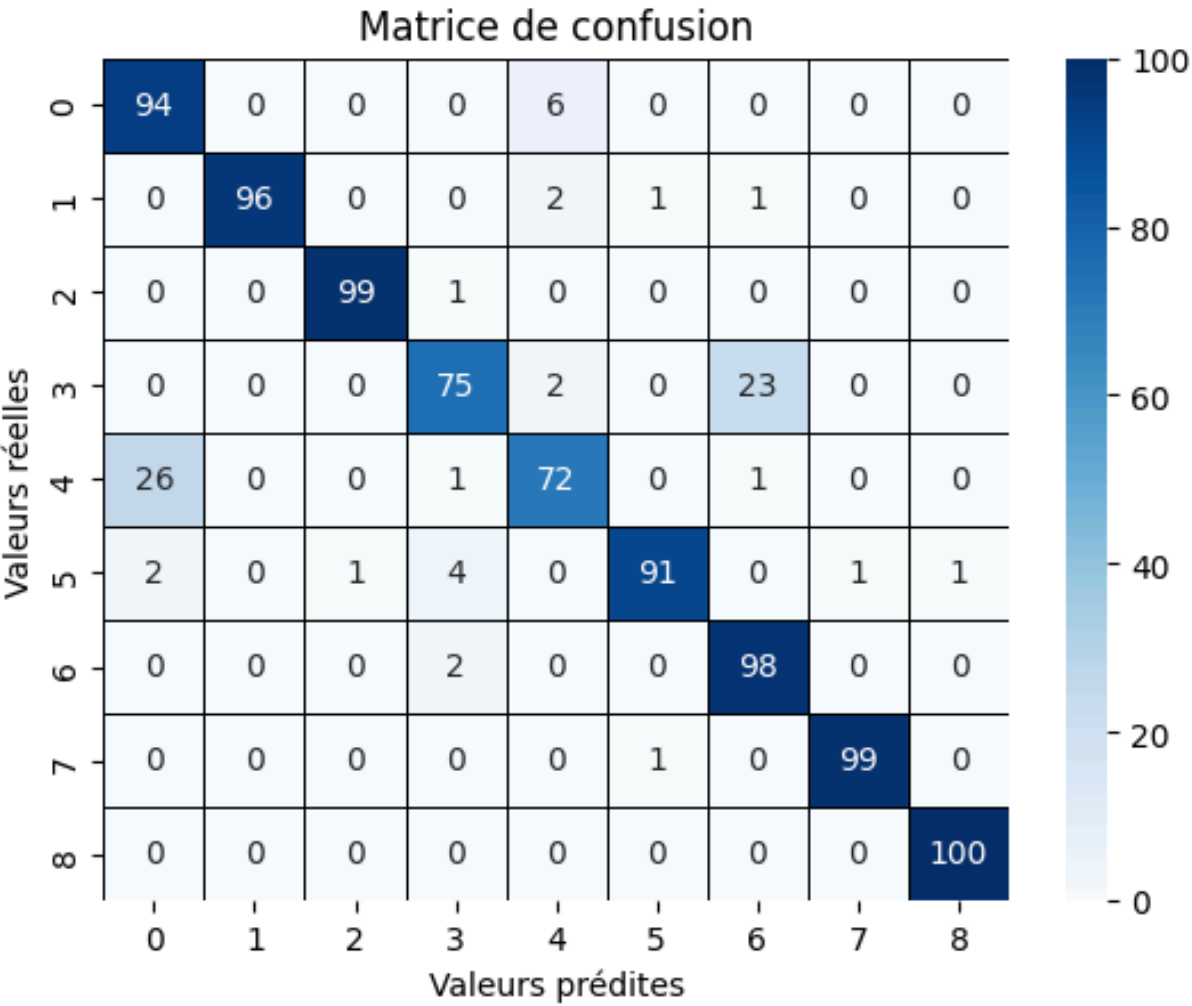
Trois canaux de couleurs (RGB)



Caracteristiques extraites



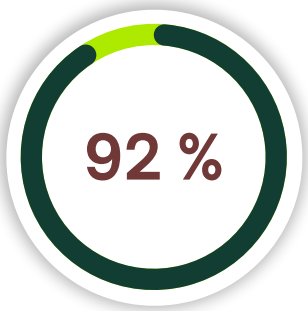
Matrice de confusion



Analyse des resultats

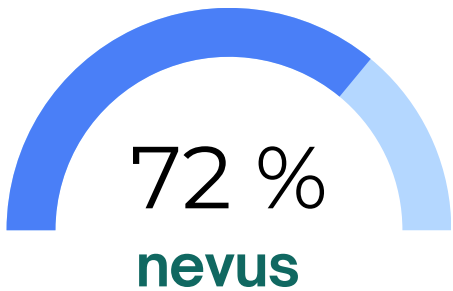
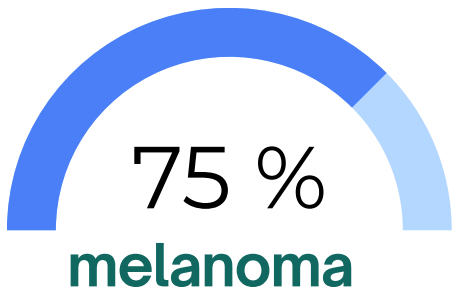
Rapport de classification

	precision	recall	f1-score	support
0	0.77	0.94	0.85	100
1	1.00	0.96	0.98	100
2	0.99	0.99	0.99	100
3	0.90	0.75	0.82	100
4	0.88	0.72	0.79	100
5	0.98	0.91	0.94	100
6	0.80	0.98	0.88	100
7	0.99	0.99	0.99	100
8	0.99	1.00	1.00	100
accuracy			0.92	900
macro avg	0.92	0.92	0.91	900
weighted avg	0.92	0.92	0.91	900

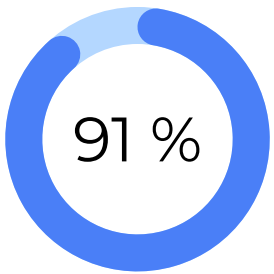


Précision globale

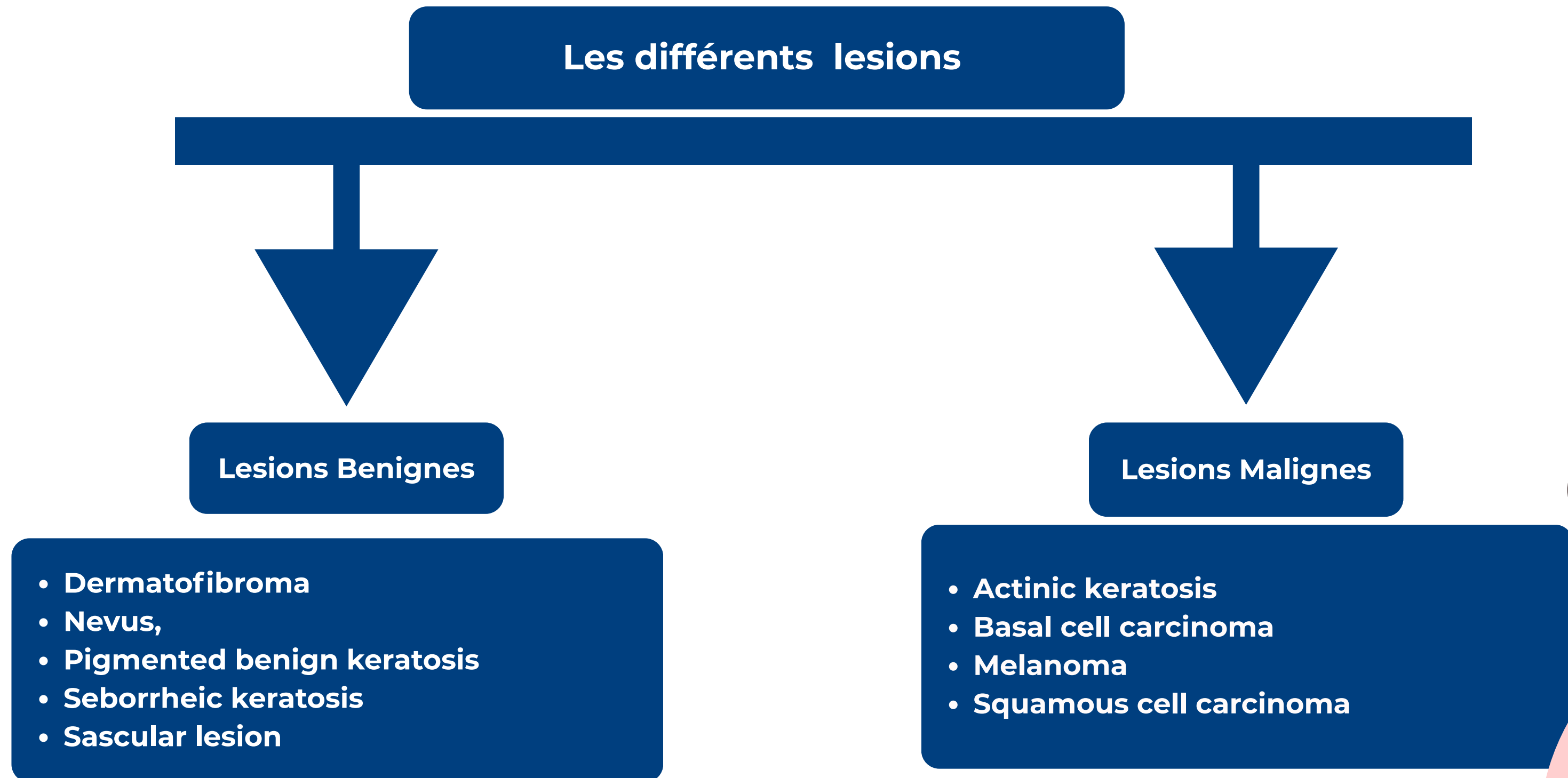
Recall



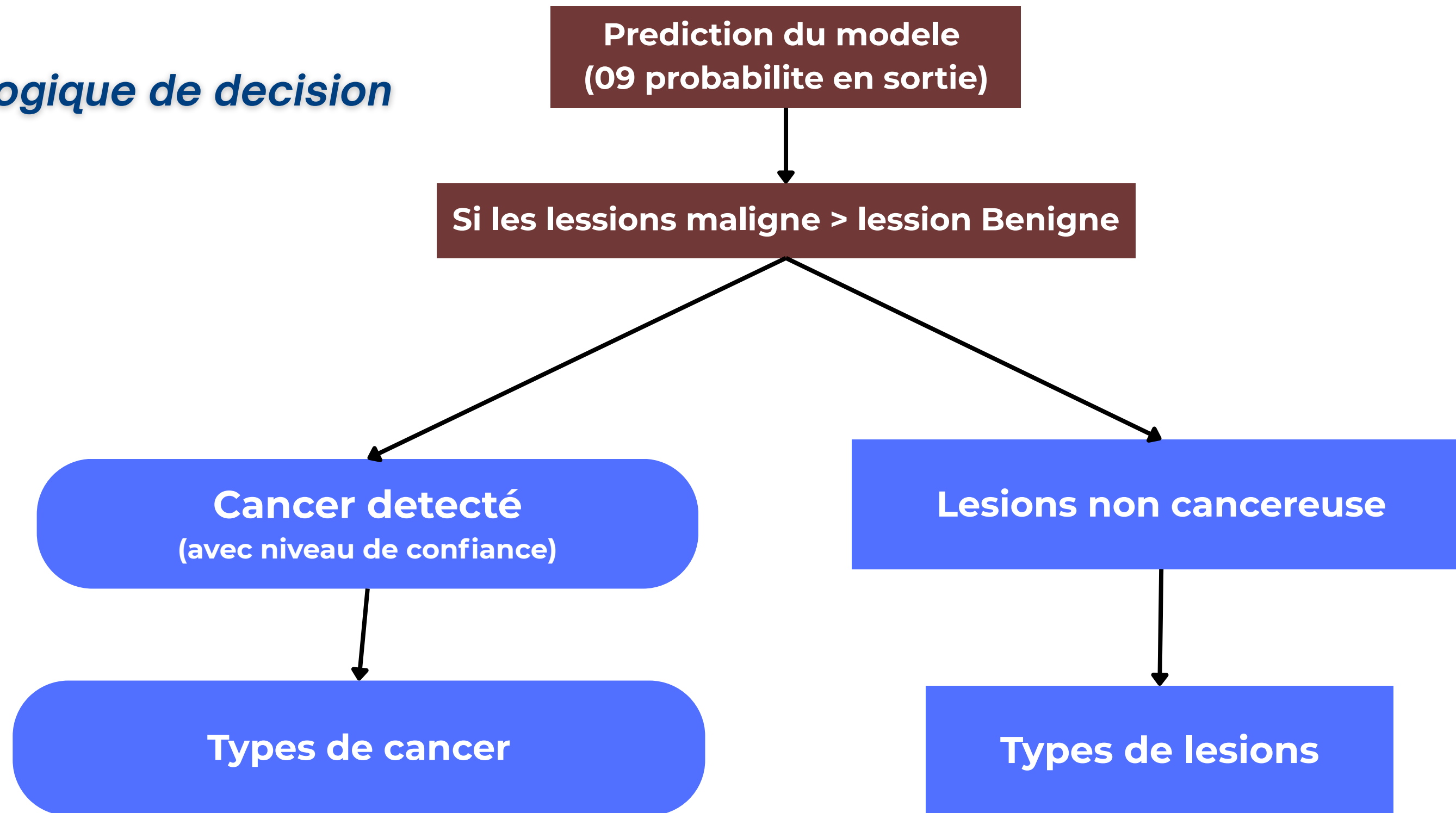
Recall moyen classes malines

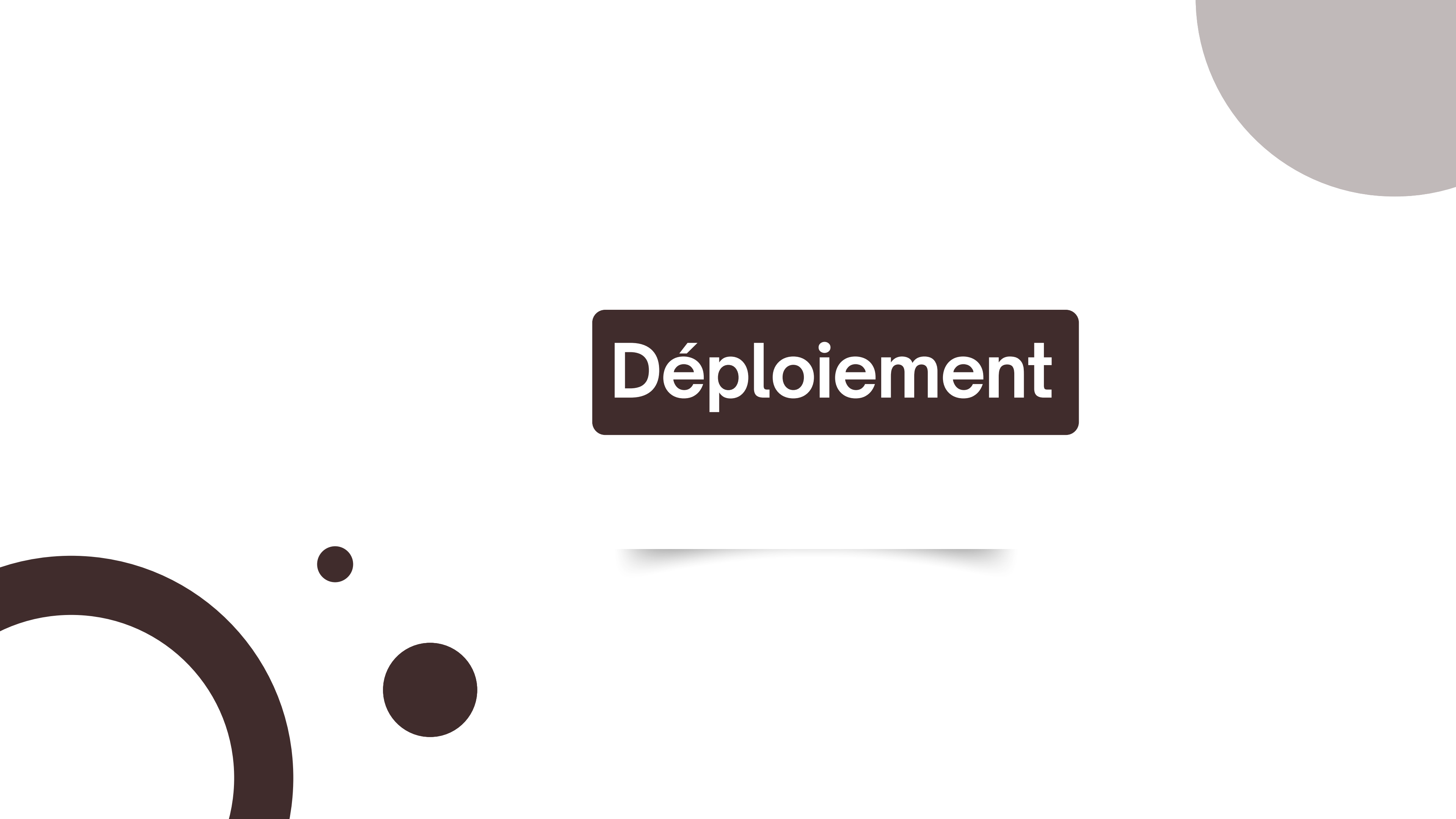


Logique de Decision



Logique de Decision





Déploiement